ME3 Voice Duplex: A Tool-Using, Full-Duplex Voice Agent for Raspberry Pi

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Abstract

ME3 is a Raspberry Pi voice-agent prototype that combines local speech processing, full-duplex audio, a remote Qwen language-model brain, and project-owned Model Context Protocol (MCP) services. The system is designed for an important interaction beyond question answering: a user can ask by voice for a web application or other coding task, and the model can call bounded file, search, command, and background-job tools to perform the work. No task-specific model training is required. Instead, the system integrates pretrained Silero VAD, faster-whisper ASR, Piper TTS with an optional streaming ElevenLabs voice, and an OpenAI-compatible Qwen/vLLM endpoint. This paper describes the architecture, tool-call protocol, safety boundary, local verification, Raspberry Pi release workflow, and evidence boundary for the separate FULL-DUPLEX-BENCH v1.5 evaluation. The current implementation is an engineering release, not a claim of completed paper-grade overlap results. The current revision also specifies the Vocalis-Pi event contract: continuous VAD/transcript inputs, explicit turn states, interrupt controls, and an adapter seam for asynchronous MCP harness work. The workstation acceptance passed 36 tests and produced a 52-file prepared release; the last Pi receipt still refers to an earlier deployed release, so this revision is not presented as a fresh Pi deployment.

Keywords

voice agents, full duplex, Raspberry Pi, tool calling, MCP, Qwen, speech interaction, edge computing

1 Introduction

Voice interfaces are often presented as a sequence: wait for the user to stop, transcribe one utterance, generate text, and play a response. A useful assistant must also manage interruption and action. A user should be able to say "build a web app" and receive evidence of files and verification, rather than a plausible paragraph describing code that was never created.

ME3 addresses these requirements with a local audio loop and a remote tool-using language model. The design separates four concerns. First, the Pi keeps voice-sensitive processing local through VAD, ASR, and TTS. Second, a Qwen model served by an OpenAI-compatible vLLM API performs dialogue reasoning. Third, typed tools provide an auditable action surface. Fourth, a release and deployment workflow distinguishes local preparation from actual Pi installation.

The project is intentionally not a new model-training project. The agent behavior comes from inference and orchestration: pretrained speech models, Qwen inference, MCP/tool dispatch, and a bounded local execution policy. This choice reduces iteration time and makes the engineering questions explicit.

The contributions of this work are:

- a full-duplex local voice loop with barge-in during TTS playback;
- an OpenAI-compatible Qwen tool loop that can create and verify a static web application through voice or text;
- one bounded registry shared by direct Qwen calls and MCP wrappers;
- confirmation-gated Twilio call/SMS and Resend email adapters with a safe provider-response boundary;
- background task supervision so builds do not block the microphone;
- a reproducible local-dev, package, deploy, and evidence workflow for the Raspberry Pi target.

2 System Requirements and Scope

The required runtime contains four owned MCP services: VAD, ASR, TTS, and tasks. Pipecat is retained as the harness-facing integration path, while the local loop is the reference implementation used for Windows development and the Pi service. The runtime configuration is kept in one YAML file so the audio rate, speech thresholds, model choices, endpoint, tool budget, and deployment path are not duplicated across scripts.

The current engineering scope includes interruptible spoken conversation, remote Qwen inference, project-local coding tools, public web lookup, timers, system information, the existing ME2 bridge, and credential-gated outbound call, SMS, and email actions. It does not claim unrestricted Onit equivalence. An always-listening device needs a narrower default action surface than an interactive coding IDE. Capabilities that would add external side effects, credential use, repository rewrites, or arbitrary shell access must be added as reviewed typed tools.

3 Architecture

3.1 Audio and conversation path

The audio path is:

`microphone -> Silero VAD -> faster-whisper -> Qwen -> Piper/ElevenLabs -> speaker`.

Capture remains active while TTS plays. The local state machine has four states: `idle`, `listening`, `thinking`, and `speaking`. Speech during `speaking` stops playback and starts a new turn. A short pre-roll is retained so the beginning of an utterance is not clipped. The VAD endpoint requires a minimum speech interval followed by configured trailing silence.

When `ELEVENLABS_API_KEY` and `ELEVENLABS_VOICE_ID` are provisioned, the TTS server streams raw 16 kHz PCM from the ElevenLabs low-latency endpoint. Otherwise the same path uses the local Piper voice. The Philippines-market profile sets `language_code=fil` for Filipino and keeps English and Taglish in the supported interaction set. A matching ElevenLabs voice ID must be selected and provisioned by the operator; the repository does not embed a voice identity or credential. Qwen remains the reasoning and tool-calling model. ElevenLabs Realtime v1 Mini/Max is a separate hosted voice-agent architecture that also manages the LLM and ASR settings, so it is not silently substituted into this experiment.

On 26 October 2026, an authorized local provider-only test used the gitignored Windows development credential and an operator-selected account voice. ElevenLabs returned 54 raw-PCM chunks (163,468 bytes at 16 kHz), and a Filipino-language test phrase played through the workstation's default speaker. The adapter was corrected to reassemble 16-bit samples split across HTTP chunk boundaries and to classify chunked-encoding failures using the correct Requests exception handling. The account voices inspected for this test were English-labelled; therefore this result verifies the provider integration and output path, not approval of a native Filipino accent.

The duplex state machine is independent of the Qwen model. This makes the interaction control testable without a network request and leaves room for a Pipecat transport to reuse the same transition logic.

3.2 Vocalis-Pi event contract

The implementation makes the full-duplex behavior observable through a transport-independent Vocalis-Pi orchestrator. Its external states are `LISTENING`, `THINKING`, `SPEAKING`, `BACKCHANNELING`, and `INTERRUPTED`. The local loop remains active on the microphone while the speaker plays. A VAD update received during `SPEAKING`, `THINKING`, or `BACKCHANNELING` publishes a `barge_in_event`, then emits `FLUSH_QUEUE` and `FADE_OUT` TTS controls before returning to `LISTENING`.

The protocol carries four input/event types: VAD state, transcript deltas, overlap signals, and asynchronous harness events. Its output types are assistant-text deltas, TTS control actions, turn-state transitions, and optional harness dispatch requests. A long Qwen tool turn can therefore produce a short `BACKCHANNELING` acknowledgement while its task continues. Tool progress is mirrored as sanitized harness events; arguments and results are excluded from these events because they can contain user data or credentials. The exact channel names and JSON examples are documented in the repository's Vocalis protocol manual.

The optional harness adapter emits a JSON-RPC 2.0 `tools/call` request named `harness_dispatch`. The default verified release keeps the bounded Qwen tool loop in-process for low overhead and uses the adapter as a stable seam for an Onit, Pipecat, or OpenCode background worker. This makes the contract testable without claiming that a separate external harness is already running in every deployment.

3.3 Tool-using Qwen brain

The remote brain is an OpenAI-compatible chat-completions client. The first request contains the recent dialogue and JSON Schema definitions for the ME3 tools. Qwen may return assistant text or a list of function calls. For every call, ME3 parses the arguments, executes the local function, appends a `role=tool` message, and asks Qwen to continue. The maximum number of tool rounds is eight by default; the checked-in configuration uses ten.

The protocol is illustrated below.

```
{
  "model": "Qwen/Qwen3.8-27B",
  "messages": [{"role":"user", "content":"build a web app"}],
  "tools": [{"type":"function", "function":
    {"name":"write_file", "parameters": {}}}],
  "tool_choice": "auto"
}
```

After a tool call, the result is returned as structured JSON. Qwen therefore receives the actual file path, byte count, return code, or error. The final spoken response is not accepted as evidence by itself.

3.4 Owned MCP services

The VAD server exposes frame and stream detection plus endpointing. The ASR server exposes PCM and file transcription. The TTS server exposes synthesis, streaming chunks, and installed voices. The tasks server retains timers, system information, and the ME2 bridge, and now wraps the same bounded project tool registry used by the direct Qwen path. The shared implementation avoids one policy for the local loop and another policy for MCP clients.

4 Tool Registry and Safety Boundary

Table 1 summarizes the direct functions visible to Qwen.

The path policy resolves every path inside the ME3 project. Credential-like names, environment files, private keys, the Git directory, the virtual environment, model caches, and generated release trees are protected. Reads and writes are size bounded. Commands are tokenized and run without a shell; shell operators and patterns associated with deletion, privilege escalation, SSH/SCP, repository reset/clean/push, and package publication are rejected. Long commands run with standard input closed, captured output, and a timeout.

This boundary is a design choice rather than an implementation accident. A voice transcript can be ambiguous, and the assistant is not an appropriate place for silent arbitrary shell execution. The registry is sufficient for the demonstration web app while keeping higher-risk operations explicit.

Table 1: Direct Qwen tools in the ME3 registry

Tool	Purpose
`list_files`	List project files and sizes.
`read_file`	Read bounded project text.
`write_file`	Create or replace a project text file.
`search_text`	Search project text files.
`run_command`	Run a direct allowlisted command.
`start_background_task`	Start a bounded build or test.
`task_status/output/cancel`	Supervise a background task.
`system_info`	Return non-secret host information.
`fetch_url/web_search`	Read public HTTP(S) resources.
`git_status`	Inspect repository status without mutation.
`channel_health`	Report safe call, SMS, and email configuration.
`send_sms/make_call/send_email`	Confirmation-gated external communication through Twilio and Resend.

The external communication tools use the same registry and are stricter than the local project tools. `channel_health` is network-free. Each `send_sms`, `make_call`, and `send_email` request first validates its target and returns a confirmation-required status unless the operator has explicitly confirmed the side effect. The adapters then send one bounded REST request and return only a status, channel, and safe provider ID; provider response bodies and authorization headers are discarded. Twilio call initiation additionally requires a reviewed public HTTPS TwiML URL. A complete call-center bridge, inbound webhooks, identity/CRM actions, consent, recording, and human transfer remain outside this adapter boundary.

5 Web-App Capability

The target workflow is a dependency-free web application when the user does not name a framework. Qwen writes an application directory under `agent_workspace/<name>`, lists the result, reads the important files, and runs a bounded verification command if one is appropriate. The provided demonstration contains `index.html`, `styles.css`, and `app.js`; it provides a command input, microphone placeholder, status area, and tool-activity log.

The capability is demonstrated twice: the first live request used Qwen to create the HTML file through a real tool call; a later live request used Qwen to list the directory and read all three files. This demonstrates inference, tool dispatch, file mutation, and evidence retrieval without training a model. The demo is intentionally local and static; the current safety policy does not publish it to the public internet or install an arbitrary server.

6 Deployment Workflow

ME3 follows a four-stage local-dev workflow:

(1) **Environment gate:** import the shared Python dependencies.
(2) **Smoke gate:** load all four MCP services, call their health tools, construct the engineering benchmark, and query PortAudio when available.
(3) **Package:** copy the runtime files into a versioned release, write SHA-256 entries, and create a zip archive.
(4) **Deploy:** transfer the archive to the configured Pi, merge its stable `me3_voice_duplex` directory, run the idempotent installer, and record a deployment receipt.

The current target is dalmacio@cfdfnjrpi5.local at /home/dalmacio/Desktop/kiko/me3_voice_duplex. The separate ME2 deployment remains at /home/dalmacio/Desktop/dandan. A prepared release is not called deployed. The local receipt must say `deployed` after SSH, SCP, extraction, and installation succeed. The current receipt reports `deployed`; after installation, a `--check` pass and authenticated Qwen model/chat probes completed on the Pi. The read-only Fifine preflight currently reports no PortAudio capture device, so the Pi production user service remains disabled. HDMI playback is enumerated, but no physical Pi speaker turn or human spoken turn is claimed. At this revision, the receipt's deployed identifier is an earlier release; the new 52-file Vocalis release is locally prepared and has not been transferred.

The API key is never included in the archive. On the Pi it must be provisioned through a secure runtime environment. The Windows DPAPI helper can report configured/not-configured and can provide an in-memory child-process value for local testing, but its secret is not printed or written to the repository.

7 Verification and Evidence

The current local verification sequence is:

```
python scripts/dev_env_check.py
python scripts/local_smoke_test.py --with-audio
python scripts/check_paper_alignment.py
python -m pytest tests -q
python -m me3.agent --brain remote --text \\
  "Use system_info and summarize the host."
```

The complete test suite passed 36 tests in the current checkout. The tool-focused tests cover OpenAI-compatible schemas, blocked shell patterns, background task completion, and a mocked multi-step Qwen response. The ElevenLabs tests additionally cover split-byte PCM reassembly, genuinely odd total PCM rejection, timeout retry, and chunked-encoding retry without credential disclosure. The channel tests cover confirmation gating, E.164 validation, Twilio request construction, HTTPS call-webhook enforcement, Resend request construction, and safe metadata. Python compilation also passed. The Vocalis tests cover state transitions, JSON-RPC harness dispatch, barge-in TTS-control ordering, and LocalVoiceLoop tool-progress bridging. One current injection-only fixture run also completed with one turn and zero errors; it remains engineering evidence and not physical-speaker or v1.5 evidence. The live Qwen smoke test records an actual tool call and final

Table 2: ME3 progress evidence at the 27 October 2026 revision

Evidence lane	Status and interpretation
Tool-capable Qwen	Verified in earlier live runs: Qwen created and inspected a web-app artifact through bounded tools.
Local duplex audio	Verified with the Windows microphone, Piper output, and two barge-in cancellations; echo can contaminate ASR on loud speakers.
ElevenLabs renderer	Verified locally: provider-live 16-kHz PCM playback through the default speaker; not a native-Filipino voice-quality result.
Qwen endpoint	Authenticated model discovery and chat probe pass from the deployed Pi; a full spoken turn is still blocked by capture hardware.
External channels	Twilio/Resend adapters and fake-request tests pass; no live call, SMS, or email side effect has been placed.
Pi hardware	Deployment receipt is verified, but PortAudio reports no capture device; physical Pi speaker and human-turn evidence remain open.
Paper-native FDB v1.5	Incomplete: no clean 499-utterance overlap artifact or paper-grade result is claimed.

answer. The local smoke gate loads the four MCP servers and confirms the current Windows PortAudio inventory. These are engineering checks, not a claim about Pi audio.

The evidence status at this revision is summarized in Table 2.

The deployed Pi preflight is also an explicit hardware gate. It is read-only, does not alter mixer state, and returns exit code 2 when no capture device is visible. This keeps a playback-only Pi from being mislabeled as a working full-duplex voice endpoint.

When speaker hardware is unavailable, the project uses a separate injection-only lane. The evaluator feeds recorded 16 kHz PCM into the same VAD, ASR, Qwen/tool, TTS, and interruption state machine at wall-clock frame cadence, while a virtual sink records TTS timing without opening PortAudio output. This verifies injected-input agent behavior and virtual barge-in, but does not claim physical-speaker behavior or paper-native v1.5 evidence.

The project also contains a separate engineering latency driver and a paper-native FULL-DUPLEX-BENCH v1.5 contract. The v1.5 contract includes four scenarios with 200, 99, 100, and 100 stimuli and response categories RESPOND, RESUME, UNCERTAIN, and UNKNOWN. The engineering five-scenario driver must not be presented as the paper's 499-utterance result. A complete captured overlap artifact, transcript classification, and prosodic before/after analysis are still required before claiming a paper-grade result.

8 Discussion

The main engineering trade-off is locality versus model capability. Keeping speech I/O local reduces the need to upload audio and makes the Pi responsive, while Qwen supplies stronger reasoning and code generation than a small local model. The cost is network dependence, endpoint latency, and the need to provision the API key on the target device. The typed registry makes the boundary visible and testable, but it is not a complete replacement for an interactive coding environment.

The local ElevenLabs result narrows one implementation risk: the selected streaming endpoint can produce audio in the PCM format consumed by the full-duplex playback path, and the adapter tolerates arbitrary HTTP chunk boundaries. It does not remove the more important system risks. A complete voice-agent turn still requires both a reachable Qwen brain and a voice approved for the intended Filipino customer-support deployment. The new communication adapters narrow the integration boundary but do not establish that the agent can safely replace a staffed call center: provider accounts, webhooks, media streaming, consent, retention, escalation, and supervised quality gates remain necessary. Provider rendering, Qwen/tool execution, and external-channel side effects are therefore reported as separate evidence lanes rather than collapsed into one end-to-end claim.

The full-duplex behavior is also more than a transport setting. VAD must keep running during playback, the audio output must be cancellable, and the turn-state transition must preserve the interrupted user's utterance. Echo control, microphone placement, and ALSA device selection remain physical Pi concerns. A successful Windows loopback does not prove those conditions.

Future work should add a Pi-local live self-test, a reviewed HTTPS call/media-stream bridge, inbound webhook signature verification, secure credential provisioning and rotation, operator-owned Twilio/Resend tests, a reviewed web-app preview server with explicit approval, and the full v1.5 overlap capture/evaluation. Additional Onit-like capabilities should be introduced as typed, auditable functions rather than by removing the voice safety boundary.

9 Conclusion

ME3 now combines full-duplex voice interaction with a real Qwen tool loop and an explicit Vocalis-Pi event contract. A spoken build request can result in actual project files and verification, and the agent can supervise longer work without freezing the microphone. The implementation uses inference and tools rather than model training, keeps local speech services under project ownership, and records the distinction between local verification, prepared release, Pi deployment, and benchmark evidence. A provider-live ElevenLabs rendering now plays through the local Windows speaker, but the account voices inspected for that test were English-labelled and an approved Filipino/Philippine-English voice is still required. The current workstation release passes 36 tests and the injection lane passes one clean fixture run; the latest

prepared bundle is not yet the Pi-installed bundle. The next decisive checks are a capture device, a Qwen-plus-voice turn, operator-owned live channel tests, and a clean captured v1.5 overlap evaluation.

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